Week 1 of training

Date- 3 july to 10 july

Day 1:-

Objective

Understand the scope of the six-month industrial training, software development lifecycle, and set up the development environment.

Topics Covered

- **Introduction to Full Stack Development**
- **Roles of Frontend, Backend, Database**
- **Overview of Python, JavaScript, AI, Mobile Development**
- **Software Development Life Cycle (SDLC)**
- **Agile Development Methodology**
- **Industry Coding Standards**

Practical Work

- **Installed Visual Studio Code**
- **Installed Python and configured environment variables**
- **Installed Node.js and npm**
- **Installed Git**
- **Created GitHub account/repository**
- **Configured Git using terminal**

Day 2:-

Objective

Build a strong understanding of Python programming.

Topics Covered

- **Python Syntax**
- **Variables**
- **Data Types**
- **User Input**
- **Type Casting**
- **Operators**
- **Comments**

Practical Work

Created multiple beginner programs:

- **Calculator**
- **Temperature Converter**

Day 3:-

Objective

Understand program flow using conditions and loops.

Topics Covered

- **if**
- **if-else**
- **Nested if**
- **for loop**
- **while loop**
- **break**
- **continue**
- **range()**

Practical Work

Implemented:

- **Prime Number Checker**
- **Even/Odd Checker**

Day 4:-

Objective

Learn reusable programming techniques.

Topics Covered

- **Functions**
- **Parameters**
- **Arguments**

Practical Work

Developed:

- **Student Grade Calculator**

Day 5:-

Objective

Understand version control and collaboration.

Topics Covered

- **Git Basics**
- **Repository**
- **Commit**
- **Branch**
- **Merge**
- **Clone**
- **Push**
- **Pull**

Practical Work

Created repository

Day 6,7:-

Objective

Revise concepts learned throughout the week and integrate them into a small project.

Topics Revised

- **Python Basics**
- **Conditional Statements**
- **Loops**
- **Functions**
- **Git & GitHub**
- **HTML FundamentalsMini Project**

Student Information Management System

Features

- **Add Student Details**
- **Display Student Information**
- **Calculate Percentage**
- **Grade Generation**
- **Store Data Using Lists**

Technologies Used

- **Python**
- **Git**
- **GitHub**
- **HTML**